MA'ADEN ENGINEERING MANUAL

Roadworks

MD-101-SMEM-EG-CV-SPC-0003

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for Roadworks for all Ma'aden projects. It does not cover concrete slabs or concrete paved roads, and bridgeworks, but includes minor concrete works as in Section 5.13. Geotechnical investigation is not part of the scope of this document.

This document is intended to be used in conjunction with the referenced documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for Roadworks for all Ma'aden projects.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Site	Geographical location where Services are to be performed.
Wearing Surface (Surface Course)	The top layer of a Pavement.
Graded Base Course	The layer, or layers, of specified or selected material of designed thickness placed on a sub base or a sub-grade to support a wearing course. This layer is also called Base Course.
Asphaltic Base Course	A Base Course comprising a suitably graded Asphaltic Concrete containing asphalt cement, referred to as "AC Base Course".
Sub-base	The layer or layers of specified or selected material of designed thickness placed on a sub-grade to support a graded base course.
Subgrade	The top in-situ material (treated or untreated) of a roadbed or parking lots upon which the pavement structure and/or shoulders are constructed.
Embankment	A raised earth structure on which the roadway pavement structure is placed.

Asphalt	A dark brown to black cementitious material; solid, semisolid, or liquid in consistency; in which the predominating constituents are bitumens which occur in nature as such or which are obtained as residue in refining petroleum.
Asphalt Cement	A fluxed or unfluxed asphalt especially prepared as to quality and consistency for direct use in bituminous pavements, and having a penetration at 25 C of between 5 and 300 under a load of 100 g applied for 5 s.
Asphaltic/Bituminous Concrete	A combination of mineral aggregates and asphalt cement, mixed in a central plant which is usually mixed, laid, and compacted while hot.
Bitumen	(1) A hydrocarbon material, obtained from natural asphalts or refined petroleum, which is completely soluble in carbon disulfide. (2) Bitumen is a common synonym for asphalt cement.
Bituminous Pavement	A pavement composed of crushed rock or other aggregate cemented together with bitumen.
Surface Treatment	A sprayed-on application of bituminous material, with or without a thin layer of covering aggregate.
Seal Coats	Surface treatments that have waterproofing or texture improvement, or both, as their main purpose.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards or codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to the following.

3.2.1 Local Codes and Standards

3.2.1.1 Ministry of Transportation

- General Specification for Roads and Bridges
- Highway Materials Manual

3.2.2 International Codes and Standards

3.2.2.1 American Society for Testing and Materials (ASTM)

ASTM C33	Standard Specification for Concrete Aggregates
ASTM C88	Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C131	Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C150	Standard Specification for Portland Cement
ASTM C171	Standard Specification for Sheet Materials for Curing Concrete
ASTM C260	Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C309	Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM D8	Standard Terminology Relating to Materials for Roads and Pavements
ASTM D242	Standard Specification for Mineral Filler for Bituminous Paving Mixtures
ASTM D 422	Standard Method for Particle-Size Analysis of Soils
ASTM D692	Standard Specification for Coarse Aggregate for Bituminous Paving Mixtures
ASTM D977	Standard Specification for Emulsified Asphalt
ASTM D994	Standard Specification for Preformed Expansion Joint Filler for Concrete (Bituminous Type)
ASTM D1073	Standard Specification for Fine Aggregate for Bituminous Paving Mixtures
ASTM D1190	Standard Specification for Concrete Joint Sealer, Hot-Poured Elastic Type
ASTM D1556	Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method

ASTM D1557	Standard Specification for Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort
ASTM D1559	Standard test method for Resistance to Plastic Flow of Bituminous Mixtures using Marshall Apparatus
ASTM D1751	Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Non-Extruding and Resilient Bituminous Types)
ASTM D1883	Standard Test Method for CBR (California Bearing Ratio) of Laboratory Compacted Soils
ASTM D2026	Standard Specification for Cutback Asphalt (Slow-Curing Type)
ASTM D2027	Standard Specification for Cutback Asphalt (Medium-Curing Type)
ASTM D2028	Standard Specification for Cutback Asphalt (Rapid-Curing Type)
ASTM D2041	Standard Test Method for Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures
ASTM D2172	Standard Test Methods for Quantitative Extraction of Bitumen From Bituminous Paving Mixtures
ASTM D2487	Standard Specification for Classification of Soils for Engineering Purposes
ASTM D2628	Standard Specification for Preformed Polychloroprene Elastomeric Joint Seals for Concrete Pavements
ASTM D2726	Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures
ASTM D2940	Standard Specification for Graded Aggregate Material for Bases or Sub-Bases for Highways or Airports
ASTM D3282	Standard Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes
ASTM D3381	Standard Specification for Viscosity-Graded Asphalt Cement for Use in Pavement Construction
ASTM D3628	Standard Practice for Selection and Use of Emulsified Asphalts
ASTM D4253	Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

ASTM D5361	Standard Practice for Sampling Compacted Bituminous Mixtures for Laboratory Testing
ASTM D5821	Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate
ASTM D6938	Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

3.2.2.2 American Association of State Highway and Transportation Officials (AASHTO)

AASHTO MAPH-2	Hot-Mix Asphalt Paving Handbook
AASHTO TP61-02	Standard Method of Test for Determining the Percentage of Fracture in Coarse Aggregate
AASHTO T27	Standard Method of Test for Sieve Analysis of Fine and Coarse Aggregate
AASHTO T11	Standard Method of Test for Materials Finer than 75 um (No. 200) Sieve in Mineral Aggregates by Washing
AASHTO M-180	Standard Specification for Corrugated Sheet Steel Beams for Highway Guardrail
AASHTO M133	Standard Specification for Preservatives and Pressure Treatment Process for Timber

3.2.2.3 American Concrete Institute (ACI)

ACI 304R-00	Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 301-10	Specifications for Structural Concrete

3.2.2.4 US Department of Transportation

Manual on Uniform Traffic Control Devices (MUTCD)

3.2.3 Alternative Design Codes

Alternative codes may be used for Roadworks if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0004	Stormwater Drainage and Underground Gravity Sewers
MD-101-SMEM-EG-CV-SPC-0009	Geotechnical Investigation
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

3.4 Site Conditions

All materials, equipment and Work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. MATERIALS

4.1 General Requirements

The construction of all roadworks shall conform to the relevant standards, governing authorities and this specification.

Surface treatment shall only be used on straight sections of roads not subject to plant maintenance activities and/or trucks turning.

All pavements around process plants, plus intersections and turns of other roads shall either be surfaced with hot-laid asphaltic concrete, or shall have concrete pavements.

4.2 Material Specifications

All material used for constructing roads shall be free of roots, foreign material and any other deleterious substances. The Contractor shall comply with the following:

- Standard Terminology as in ASTM D8 - 02 shall be followed
- General requirements for soil and aggregate classification shall conform to ASTM D2487, ASTM D2940 and ASTM D3282
- Portland cement shall be Type I and Type V as per ASTM C150
- Air-entraining mixture shall conform to ASTM C260
- Preformed expansion joint fillers shall conform to ASTM D994, ASTM D1751 or ASTM D2628
- Poured joint sealant shall conform to ASTM D1190
- Waterproof paper shall conform to ASTM C171
- Burlap shall be clean and free of substances that may be harmful to freshly laid concrete
- Polyethylene sheeting shall conform to ASTM C171
- Curing compound shall conform to ASTM C309.

4.3 Select Fill (Class 1)

Materials shall conform to the following requirements in addition to those specified for "Common Fill (Class 2)" material:

- The Plasticity Index shall be less than 6 and liquid limit less than 25, when tested in accordance with ASTM D4318
- The CBR of the material shall be a minimum of 15% or as specified by the project
- The weight of soluble salts in soil shall be less than 3%
- When tested in accordance with ASTM D422, the material shall comply with the grading requirements nominated in Table 4-1.

Table 4-1 : Fill Grading Requirements

Sieve Size	% Passing
51.0	100
37.5	95 – 100
19.0	50 – 85
6.7	40 – 80
2.36	35 – 70

4.4 Common Fill (Class 2)

Common Fill (Class 2) material shall be suitable excavated material free of deleterious substances such as expansive clay, rubbish, organic, perishable or un-compactable material, waste concrete, or other debris.

It shall be suitably graded to ensure that when compacted it derives its stability from compaction of the fine material around the coarser particles. Rock material must be broken down and evenly distributed throughout the layer to prevent the formation of voids and produce a dense compact embankment. To meet this requirement, additional fine material may need to be obtained from other sources and blended through the fill.

Unless otherwise specified in the project documentation, when tested in accordance with ASTM D422, the material shall be well graded with 100% less than 150 mm in the greatest dimension and a maximum of 15% passing the No. 200 sieve, except that the largest size shall not exceed $^1/_3$ the layer thickness after compaction.

4.5 Aggregates

Aggregate shall be composed of inert, durable material, reasonably uniform in quality and free from soft or disintegrated particles. Aggregate sub-base, base course materials and aggregate surface materials shall conform to the following requirements:

- Soundness to be tested according to ASTM C88 and the material should comply with the values given in Table 4-3 and Table 4-5
- All crushed aggregate when tested according to ASTM C136 and ASTM C117 to be well graded and conform to project specified gradation limits
- All crushed aggregate when tested according to ASTM D5821 and AASHTO TP61-02, Fracture Count for Coarse Aggregate, Method "A", shall meet the following minimum requirements:
  - Heavily trafficked roads (as defined in the project documentation) – 75% of the material retained on the 4.75 mm sieve must have one or more fractured faces
  - Other roads – 60% of the material retained on the 4.75 mm sieve must have one or more fractured faces
- The plasticity index for crushed aggregate shall not exceed 6.0.

4.5.1 Additional Requirements for Sub Base

In addition to the above, Sub base shall conform to the following requirements.

- Class A Sub base shall consist of well graded gravel with sand and silt
- Class B Sub base shall consist of a uniform mixture of gravel and/or stone fragments with sand, silt, and clay
- Gradation, CBR and other limits shall be as specified in Table 4-2 and Table 4-3.

Table 4-2 : Sub-Base Gradation Limits

Sieve Size	Class A Sub Base	Class B Sub Base
	Percent Passing	
62.5 mm	100	-
50 mm	90-100	100
37.5 mm	-	70-100
25 mm	-	55-85
19 mm	-	50-85
12.5 mm	-	-
9.5 mm	-	40-70
4.75 mm	35-70	30-60
2 mm	-	20-50
420 µm	-	10-30
75 µm	0-15	5-15

Table 4-3 : Sub-Base Requirements

Parameter	Class A Sub Base	Class B Sub Base
Liquid Limit (ASTM D4318)	-	25 max
Plasticity Index (ASTM D4318)	6 max	6 max
Loss by Abrasion % (ASTM C131 or ASTM C535)	50 max	50 max
Loss Using Soundness Test with Mg SO_4 (ASTM C88)	18	18
Minimum CBR at 95% Compaction	25	25

4.5.2 Additional Requirements for Graded Base Course

In addition to the above, Graded Base Course shall conform to the following requirements:

- Class A Graded Base Course shall consist of a uniform mixture of well graded aggregate
- Class B Graded Base Course shall consist of a uniform mixture of crushed rock or crushed gravel
- Gradation, CBR and other limits shall be as specified in Table 4-4 and Table 4-5.

Table 4-4 : Base Course Gradation Limits

Sieve Size	Class A Graded Base Course	Class B Graded Base Course
	Percent Passing	
62.5 mm	100	-
50 mm	90-100	-
37.5 mm	60-90	100
25 mm	42-77	60-100
19 mm	35-70	55-85
12.5 mm	26-60	-
9.5 mm	-	-
4.75 mm	15-40	35-60
2 mm	10-26	25-50
420 µm	5-15	15-30
75 µm	2-9	8-15

Table 4-5 : Base Course Requirements

	Class A Base Course	Class B Base Course
Liquid Limit (ASTM D4318)	-	25 max
Plasticity Index (ASTM D4318)	6 max	6 max
Loss by Abrasion % (ASTM C131 or ASTM C535)	40 max	40 max
Loss Using Soundness Test with Mg SO_4 (ASTM C88)	14 max	14 max
Minimum CBR at 100% Compaction	100	50

4.6 Asphalt Products

Asphalt products shall adhere to the following requirements:

- Asphalt shall be viscosity grade AC-5, AC-10 or AC-20 conforming to ASTM D3381
- Cutback asphalt shall conform to ASTM D2026, ASTM D2027 or ASTM D2028
- Emulsified asphalt shall conform to ASTM D977 and ASTM D3628
- Asphalt for prime coat shall be grade MC-30, conforming to ASTM D2027
- Asphalt cement for tack coat shall be grade SS-1H or CSS-1H emulsified asphalt, as applicable depending on properties of the substrata. It shall conform to ASTM D977 and shall be diluted as required with non-saline water.
- Asphaltic concrete for base, binder and surface courses shall be a mixture of asphalt cement, coarse aggregate, fine aggregate and mineral filler (if required). Job mix formula shall be established using the Marshall mix design method based on the parameters nominated in Table 4-6, Table 4-7 and Table 4-8.

Table 4-6 : Asphalt Concrete Requirements

Parameters	AC Base Course	Binder & Surface Courses
Flow	2.5 to 5 mm	2 to 4 mm
Stability (min.)	600 kg	800 kg
Air voids	4 to 7%	3 to 6%
Voids in mineral aggregate (min).	15%	14%

Table 4-7 : Composition of Asphalt Concrete

Sieve Size	Total Percent (by weight) Passing by Nominal Maximum Size of Aggregate			
	25.0 mm	19.0 mm	12.5 mm	9.5 mm
37.5 mm	100	– –	– –	– –
25.0 mm	90 – 100	100	– –	– –
19.0 mm	- -	90 – 100	100	– –
12.5 mm	60 – 80	– –	90 – 100	100
9.5 mm	– –	60 – 80	– –	85 – 100
4.75 mm	25 – 60	35 – 65	45 – 70	50 – 80
2.36 mm	15 – 45	20 – 50	25 – 55	40 – 54
1.18 mm	– –	– –	– –	– –
600 µm	– –	– –	– –	– –
300 µm	3 – 18	3 – 20	5 - 20	10 – 30
150 µm	– –	– –	– –	2 – 9
75 µm	2 – 9	2 – 8	3 – 7	3 - 7
Asphalt cement, percent by weight of total mix	For Base Course mixes usually in the range 3-7 For Surface Course in the range 4-7			
Typical application	Asphalt concrete base course	Asphalt concrete base course	Asphalt concrete surface course	Asphalt concrete surface course

Table 4-8 : Job Mix Formula Grading Tolerances (for Single Test)

Passing Sieve	Percent (±)
12.5 mm and larger	6
9.5 mm and 4.75 mm	5
2.36 mm and 1.18 mm	4
600 µm and 300 µm	3
150 µm	2
75 µm	1.5
Asphalt content, percent by weight of total mix	(±)0.4

4.7 Guard Rail

Guard rail beam elements including bolts and terminal sections shall be galvanized and conform to AASHTO M-180, Class A, Type1.

4.8 Concrete

Concrete and concrete reinforcement for the Works in Section 5.13 shall comply with Ma'aden Concrete Works Specifications MD-101-SMEM-EG-ST-SPC-0005.

5. EXECUTION

5.1 Site Examination and Set Out

Prior to commencing any Works the Contractor shall examine and survey the site.

All surveying shall be carried out by a licensed surveyor including, but not limited to:

- Setting out of the Works
- Recording relevant information to be included in the "As Built" drawings
- Ensuring all dimensions are in accordance with the drawings
- Installation of construction centre line pegs or suitable off set pegs which are to be maintained throughout the duration of the Works.

Offset pegs shall only be removed with written approval from the Employer.

5.2 Buried Services

Any excavations/earthworks shall be performed in accordance with Earthworks specification MD-101-SMEM-EG-CV-SPC-0001. If applicable, prior to commencing any earthworks or roadworks, an excavation permit shall be obtained from relevant authority/s.

The Contractor shall be responsible for locating any services that may be present and shall ensure that the service is clearly "marked out" prior to commencing Works.

In addition if any services are encountered that were not previously known, the Contractor shall notify the Employer immediately and include such information on the "As Built" drawings.

The Contractor shall take all the necessary precautions in order not to damage any services.

5.3 Temporary Works

The Contractor shall be responsible for all temporary Works such as by-pass roads, detours etc. and other necessary items such as warning lights, signs, fences and the like required to complete the Works in accordance with the Employer's requirements.

The Contractor shall submit details of the temporary Works to the Employer for approval, prior to commencing Works.

During the Works the Contractor shall adhere at all times to the requirements of the local authority, Ministry of Communication, the Traffic Engineering and Safety Department, Traffic Police Department and the Employer in relation to catering for traffic.

5.4 Equipment and Machinery

5.4.1 Rollers

Rollers shall either be steel wheeled power rollers or self-propelled multi-tired (rubber tired) rollers. Selection of compaction equipment is the responsibility of the Contractor, who shall ensure that it is capable of achieving the required standards of compacted density, surface shape and finish.

5.4.2 Brooms

A rotary road broom shall be capable of being rotated up to 45° from normal to the road surface.

If the rotary drawn broom cannot operate as a drag broom then a drag broom shall also be provided.

5.4.3 Bitumen Road Tanker and Site Storage Facility

The bitumen road tanker and site storage facilities shall be in good condition, be suitability lagged, fitted with heater tubes, temperature gauges and a circulation system.

In the event of a site storage facility being required the Contractor shall notify the Employer at least seven days prior to mobilizing. The notice shall also include the proposed location and layout.

5.4.4 Bitumen Sprayer

The bitumen sprayer must comply with the local authority's requirements. The sprayer shall have been tested to ensure calibration and uniformity of transverse distribution of at least spray rates 15% higher than that required. Variance in spray rate between nozzles shall not vary by more than 10% when using the type and grade of asphalt required for the Works, at the greatest shot width required.

The testing shall have been done within the last 18 months. Charts and certificates applying to such tests shall be available for the Employer to inspect if requested.

5.4.5 Aggregate Spreader

The aggregate spreader shall be capable of spreading a minimum width of aggregate of 3.7 m unless specified otherwise, and it shall be capable of spreading a uniform layer of aggregate on the sprayed binder within a tolerance of 5% of the required application rate.

5.5 Restoring Existing Pavement

5.5.1 General

The removal of the existing road seal will require that the seal is lightly scarified and graded to remove dirt and other deleterious material from the surface.

This Work shall be carried out prior to the pre-construction survey.

5.5.2 Pavement Preparation

Where required the graded base course and the sub base shall be repaired and compacted. The thickness of graded base course material layer shall be a minimum of 200 mm.

Existing entrances to parking areas, side roads and access ways shall be scarified, graded and re-compacted with the addition of sufficient base course material to provide a safe and smooth transition from the new roadworks/pavements.

5.5.3 Use of Existing Surfaces

Where an existing road surface is required for use during refurbishment/replacement, the Contractor shall maintain the road in a trafficable condition by watering, re-trimming and rolling, etc as necessary. Alternatively a bypass road shall be provided and be maintained during the Works.

5.6 Construction of New Pavements

5.6.1 General

The Contractor shall conform to Earthworks specification MD-101-SMEM-EG-CV-SPC-0001 and relevant drawings for all earthwork requirements.

The Contractor shall ensure that the minimum area is cleared to complete the Works. Any flora beyond the edge of the earthworks shall only be removed with the written approval from the Employer.

All excavations/earthworks shall be performed in accordance with Earthworks specification MD-101-SMEM-EG-CV-SPC-0001.

5.6.2 Pavements

5.6.2.1 Preparation of Sub-Grade

The Contractor shall prepare sub-grade in accordance with Earthworks specification MD-101-SMEM-EG-CV-SPC-0001.

The Contractor shall ensure that the surface upon which the pavement is to be placed has the specified surface finish and complies with the required tolerances and degree of compaction.

The Contractor shall, where directed by the Employer, rework and recompose the sub-grade for a depth of 200 mm to ensure that the specified compaction standard has been achieved prior to the commencement of pavement construction.

The Contractor shall prepare, shape, form and trim the finished sub-grade to line, grade, cross section and levels as outlined on the drawings. The sub-grade shall be compacted before surfacing operations commence.

The sub-grade shall be checked and approved by the Employer prior to the starting of sub base or base course work.

5.6.2.2 Preparation of Sub-Base and Graded Base Course

The sub base and graded base course shall each be thoroughly mixed to obtain a homogenous material, which is compacted to 95% and 98% of the Modified Proctor Dry Density respectively to within + 1% Optimum Moisture Content. Water used to achieve the correct moisture content or watering subsequent to placement, shall be of potable quality.

The thickness and dimensions of the sub base and graded base course shall be in accordance with the drawings and shall consist of specified materials as outlined in Section 4 of this specification.

Areas where the moisture exceeds the maximum allowable moisture content shall be boxed out and replaced with fresh material and shall be remixed or allowed to dry prior to compaction.

5.6.2.3 Final Preparation and Finishing

The Contractor shall not commence final preparation and profile until after the associated Works such as drainage, crossings, excavations, etc have been completed and properly backfilled and compacted.

The surface of the base course shall be finally compacted, trimmed and wet rolled so as to present a uniform tightly locked stone mosaic appearance, free from soft spots, cracks, drag marks and local hollows or high spots. The surface is then ready for brooming and sealing.

The Contractor shall provide a 3 m straight edge, templates to suit each cross section and labor so that the Employer can check the pavement profile. The surface shall not be allowed to vary by more than 15 mm from the straight edge laid parallel to the road centerline or from a template placed at right angles to the centerline of the pavement.

The cross fall of the pavement shall not exceed 5 mm/m from the design cross fall shown on the drawings.

The finished pavement levels shall be within tolerance values as in Section 6.1.1.

5.7 Base Course Salinity

Prior to applying a bituminous surfacing, the uppermost granular layer shall be tested to determine the salinity level or if any other deleterious substances are present.

If a salinity level (TDS) of higher than the value indicated by the project specification and/or a deleterious substance is detected in the graded base course, the graded base course shall be removed and replaced or suitable remedial procedures shall be submitted by the Contractor to ensure no adverse effects on the performance of the bituminous surfacing occurs.

5.8 Bituminous Surface Treatment

No asphalt spraying shall commence until the base course has been inspected and written approval obtained from the Employer together with authorization to commence Works.

The Contractor shall be responsible for the design, supply and installation of all bituminous surface treatments. The design shall be submitted to the Employer for approval.

The Contractor shall provide not less than 48 hours notice, to the Employer, of their intention to undertake priming and surfacing Works.

The application rates for the primer, binder and aggregate shall be site checked by means of a trial at no expense to the Employer. If changes are required, the proposed changes shall be submitted to the Employer for approval prior to commencing Works.

5.8.1 Base Course Surface Preparation

The Contractor shall thoroughly clean the pavement surface, immediately prior to priming or sealing, all loose material, dust and other foreign matter using an approved road broom. The sweeping shall extend 0.5 m beyond the edge of the primer and any foreign matter not dislodged by the road broom shall be removed by hand brooming.

The base course shall be allowed to 'dry back' prior to the application of the primer.

5.8.2 Preparation of Existing Sealed Surfaces

Immediately prior to applying the new seal, the existing sealed surface shall be:

- Swept free of loose stones, dust, dirt and any other foreign matter
- Inspected for defects and repaired by cleaning, priming and filling with a bituminous hot-mixed asphaltic concrete.

5.8.3 Schedule of Surfacing Works

The Contractor shall notify the Employer of the width, length and section to be sprayed at any one time prior to commencing Works.

The Contractor shall ensure that the area to be sprayed is limited to that which can be covered with aggregate at the specified rate within 10 minutes of spraying.

No asphalt spraying shall be carried out on wet or dusty pavements, while rain falls or appears imminent, during dust storms or whilst the shade air temperature or pavement temperature is below 20°C.

5.8.4 Jointing to Existing Work

A lap of at least 200 mm over the previous application shall be provided when part width asphalt spraying is to take place.

No aggregate shall be placed on the overlap strip in the previous spray run.

The spraying of the primer or binder for each run of the sprayer shall start and finish on a strip of two layered, asphalt adhered reinforced building paper laid beforehand across the entire width of the sprayed section.

The width of the paper shall not be less than 900 mm, but it shall be wide enough to ensure correct spraying rates for the full spraying length.

5.8.5 Protection of Other Works

All necessary precautions shall be taken to ensure that contamination of other areas by asphalt, asphaltic concrete or other bituminous material does not occur.

If contamination has occurred, it shall be cleaned immediately after the placement of the aggregate, and the item or area returned to its original state.

5.8.6 Primer Application

Prior to commencing spraying the Contractor shall obtain approval from the Employer.

No spraying shall be carried out whilst the pavement surface temperature is less than 20°C, wet or dusty, or when adverse weather conditions are forecast.

The base course shall be lightly sprayed with water prior to applying the primer. The base course shall be sufficiently dry to permit adherence to and penetration of the base course by the primer.

The base course surface shall be primed with the specified primer. The temperature of the primer shall be between 20°C and 80°C when applied.

Where the measured rate of application of the primer differs more than 10 % than that specified, the Work will be rejected.

At least one hour shall be allowed, after completing the spraying of the primer, prior to placing clean quartz sand to cover any free bitumen. The Contractor will then remove the sand once all the free bitumen has been "soaked up".

5.8.7 Primer Coat Cover Material in Trafficable Areas

Where the Employer requires the road not to be closed to traffic, the primer shall be covered with fine aggregate, which shall be 5 mm nominal size crushed and screened rock or sand conforming to relevant standards.

The cover material shall be embedded into the primer with a 6-10 t steel wheeled roller.

Any damage to the primed pavement coat shall be made good immediately, as the pavement must be left for at least 24 hours prior to the application of the seal coat.

5.8.8 Application Time for Surface Treatment

The surface treatment shall not be applied until at least 24 hours after the prime coat has been completed. The seal coat may be applied as a single coat or two-coat system.

5.8.9 Binder

The Contractor shall determine the grade and application rates of the binder to suit the site conditions and materials used. Adhesion and/or fluxing agent shall be applied, as necessary.

The binder shall be heated to a temperature necessary to carry out the operations of cutting (if required) and or spraying. At no time shall the bitumen suppliers recommended maximum temperature limit be exceeded.

5.8.10 Cover Aggregate Pre-Coat

The Contractor shall submit details of the pre-coating fluid for the aggregate to the Employer for approval prior to spreading.

5.8.11 Cover Aggregate

The cover aggregate shall contain no free water on the chips and shall be uniformly spread by a means of an approved type of mechanical spreader.

The Contractor shall provide details of the spreader to the Employer for approval.

The Contractor shall complete the application of the aggregate within 10 minutes of completing the spraying of the binder. The length of spray runs shall be limited to ensure compliance with this requirement.

The aggregate for single and double coat seals shall be applied at the rate which is sufficient for the binder to be totally absorbed and hence no adherence of the binder to vehicle tires under traffic.

Additional aggregate shall be applied to any bare or insufficiently covered areas.

Immediately after the satisfactory application of the aggregate for each seal coat, the surface shall be rolled with four passes of a multi tired roller.

After rolling of the surface, it shall be broomed to achieve an even distribution of aggregate. After brooming, the surface shall be rolled with a multi-tired roller for a minimum of twelve passes. Any remaining loose aggregate shall be swept off the road without causing damage to the seal.

If a two-coat seal system is required, then the second coat shall be applied on the same day unless advised otherwise by the Employer.

5.9 Primerseal

Primersealing shall be carried out in accordance with the drawings and this specification.

The primer seal shall be applied at the rates determined to suit the site conditions and materials used.

The primer seal shall be applied at a temperature between 75°C and 115°C.

The cover aggregate shall be applied, immediately after spraying the primer seal, at a rate of 1 m^3 of aggregate to a maximum of 150 m^2 of paved surface. Sufficient aggregate shall be laid to prevent lifting of the prime sealed surface.

The aggregate shall be of 7 mm nominal size.

Immediately after the application of the aggregate, the surface shall be rolled using steel wheeled rollers or multi-tired rollers. Rolling shall continue until the aggregate is embedded into the binder. Any loose aggregate shall be swept from the surface without damage to the seal.

Traffic shall not be allowed on the new Work until after sufficient rolling has taken place to ensure that aggregate, and the total mat, will not be disturbed.

5.10 Asphaltic Concrete Surfacing

5.10.1 General

Asphaltic concrete shall be placed in accordance with the drawings and with AASHTO MAPH-2.

The surface preparation shall be in accordance with this specification and AASHTO MAPH-2.

The prime coat shall be applied as specified above.

5.10.2 Mix Design

The Contractor shall be responsible for the design of the various mixes required, details of which shall be submitted for approval before any surfacing work is commenced.

The design of asphaltic concrete mixes shall conform to the parameters given in Section 4.6.

Before any asphaltic concrete surfacing work commences, the Contractor shall submit to the Employer the details of the designs of the various mixes required for the Work.

5.10.3 Setting Out and Checking Equipment

The Contractor shall provide the following items for use at the site for setting out, checking the surface levels of respective courses, checking the bituminous concrete temperature, and gauging the depth of the spread asphaltic concrete:

- Measuring rules
- Thermometers suitable for reading temperatures of mix in degrees Celsius at spreading plant and in the spread bituminous concrete
- A depth gauge graduated in millimetres
- A 3 m length straight edge constructed of materials and to an approved design and capable of being used to accurately detect surface irregularities to the limits specified.

5.10.4 Tack Coat

Before spreading the asphaltic concrete mix, the whole of the area to be sheeted shall be lightly and evenly tack coated with a fine spray of rapid breaking bitumen emulsion (refer Section 4.6).

The application rate for the tack coat shall be 0.15 l/m^2 of residual bitumen.

During the application of the tack coat, particular care shall be taken to avoid over spraying adjoining concrete surfaces including paths, Curbs and gutters. Where this has occurred, the Contractor shall be responsible for removing the sprayed material.

The tack coat shall be allowed to break (water separating from bitumen) before the mix is spread.

5.10.5 Placement and Compaction of Asphaltic Concrete

The asphaltic concrete mix shall be placed as soon as the prime/primer sealed surface has cured. In the event that the placement is delayed beyond the curing period of the prime/primer seal or if the primer sealed surface is not clean, a tack coat of primer may be applied in accordance with AASHTO MAPH-2 if requested by the Employer.

The spreading and compaction of the asphaltic concrete shall be in accordance with the following and AASHTO MAPH-2:

- Placement shall only be carried out during fine and dry conditions. No asphaltic concrete shall be placed during wet conditions, if inclement weather is forecast or the ambient temperature falls below 10°C.
- The asphaltic concrete shall be placed at a minimum temperature of 140°C immediately on delivery to the spreader
- All compaction shall be completed before the temperature of the mixture drops below 100°C
- The asphaltic concrete shall be placed in one layer of no less than 25 mm thickness unless noted otherwise on the drawings
- Spreading is to be by an approved self-propelled paving machine. Hand spreading shall only be allowed in inaccessible areas.
- Automatic level control is generally not required except at intersections or as directed by the Employer
- The compaction shall initially be by a smooth steel drum tandem roller consisting of at least two passes. Not less than twelve passes shall then follow the initial compaction from a self-propelled multi-tired roller of mass not exceeding 12T, whilst the asphaltic concrete is still hot.
- The final compaction shall be achieved using a smooth steel drum roller with a maximum of two passes. The asphaltic concrete shall be sufficiently warm to produce a smooth, dense surface. The drum and wheels of rollers shall be kept wet at all times during the rolling operations.

- If the asphaltic concrete commences to shove or crack, then rolling shall cease immediately. The Contractor shall determine the cause and advise of the corrective actions to the approval of the Employer.
- Any joints required, shall be smooth, even and waterproofed
- The density of the compacted asphaltic concrete shall not be less than 97% of the Marshall Maximum Density.

5.10.6 Finished Profile

The finished profile of the asphaltic concrete surface shall be in accordance with the following specifications and AASHTO MAPH-2.

- The surface shall be smooth, tightly bound, uniform and free from any defects
- The surface shall not vary by more than 10mm from the specified level
- No point on the surface shall differ by more than 5 mm from a 3 m straight edge laid in any direction on the surface, except for the crowned section.

5.11 Unsealed Roads

Pavement depth and width shall be designed as given on drawings. The gravel wearing course of an unsealed road shall comply with the properties as given in Section 4.8.

5.12 Roadworks Drainage

Drainage works shall be in accordance with Ma'aden specification MD-101-SMEM-EG-CV-SPC-0004 Stormwater Drainage and Underground Gravity Sewers.

The whole of the Works shall be finished to ensure that no areas will retain water and be free drainage with respect to stormwater run-off.

5.13 Concrete Curbs, Curbs and Channels, Invert Channels and Crossovers

5.13.1 General

All concrete Works shall be in accordance with Ma'aden Specification MD-101-SMEM-EG-ST-SPC-0005 Concrete Works.

The concrete shall conform to ACI 301-10 and ACI 304R-00, and shall also adhere to the following requirements:

- Specific slump of 70 mm
- Concrete for Curbing shall have a minimum compressive strength of 25 MPa
- Extruded concrete Curbing shall have a maximum aggregate size of 10 mm.

The exposed surfaces shall be finished to a steel float unpolished finish.

5.13.2 Extruded Curbing

Concrete curbing shall be cast by extrusion forming with the use of an extrusion machine, which is fitted with an automatic level control that will adjust the position of the forming tube whilst in operation.

The curbs shall have a well-compacted mass with a surface free from voids and honeycombs. The curbs shall be trowelled to a smooth finish immediately after extruding.

5.13.3 Cast In-Situ Curbing

Any curbing such as plain curbing, curb and channel, invert channels, reinforced curbing and crossovers that cannot be cast by extrusion forming shall be cast in-situ.

The formwork on the exposed Curb face shall be removed whilst the concrete is still green so as to be finished to a trowelled smooth finish.

5.13.4 Concrete Jointing

Joints in concrete curbs, curbs and channels, invert channels and crossovers shall be placed in accordance with the drawings and this specification.

5.13.4.1 Contraction Joints

Contraction joints in concrete curbs, curbs and channels, invert channels and crossovers shall be placed by sawing the concrete. The Contractor shall ensure that contraction joints are placed in accordance with the following requirements:

Plain Concrete:

- Where an expansion joint and a contraction joint coincide, the expansion shall be provided
- Spacing between joints shall be no more than 2.5 m intervals
- Contraction joints shall be provided at entry points and tangent points on curved sections
- Contraction joints shall be a 10 mm sawn groove.

Reinforced Concrete:

- Spacing between contraction joints shall be no more than 5.0 m intervals
- Contraction joints shall be provided at tangent points on curves and at any change in section size or shape
- Reinforcement shall be continuous through the contraction joint
- The joint shall be a 10 mm sawn groove.

5.13.4.2 Expansion Joints

Expansion joints in concrete curbs, curbs and channels, invert channels and crossovers shall be placed by sawing the concrete to provide the specified gap width. The Contractor shall ensure that the joints are placed in accordance with the drawings and the following requirements:

- Expansion joints shall be located at no more than 7.5 m intervals
- The expansion joint shall be a 10 mm gap for the entire depth of the concrete pavement unless noted otherwise

- In reinforced sections the joint shall be complete with suitable dowel bars and sealed with a flexible joint sealant. The dowel bar shall be greased on one side of the joint and a cap on the greased side
- The joint shall be sealed with mastic which has a bitumen impregnated caneite-backing sheet that has been approved by the Employer.

5.13.4.3 Construction Joints

Where practical the Contractor shall use an expansion or contraction joint as a construction joint.

Where contraction joints are to be used as pour breaks or construction joints for reinforced concrete sections, the reinforcement shall be lapped.

When pouring new extruded Curbing the first 150 mm of the new pour shall be cut away and removed. The gap between old and new work shall be filled by hand placing, rodding and shaping of the concrete until a satisfactory shape and finish has been achieved.

5.13.4.4 Sawn Joints

Saw cuts in concrete for contraction joints and at least one side of the expansion joint shall be installed within 24 hours of placing the concrete for that section.

5.14 Footpaths

5.14.1 General Requirements

Footpaths shall be constructed in accordance with the drawings and this specification.

In general footpaths shall be cast in-situ concrete, sealed bitumen or as specified otherwise and shall not be constructed until all drainage, sewerage, other services, verge clearing and Curbing have been completed.

They shall be constructed with a cross fall of 2% towards the Curb and the nominal width shall be 1.2 m unless noted otherwise. Where the path is to be constructed in a road reserve the path shall have the same longitudinal profile as that of the Curb, otherwise it shall generally conform to the surrounding finished ground level and or as shown on the drawings.

All earthworks for constructing footpaths shall be carried out in accordance with Ma'aden Earthworks specification MD-101-SMEM-EG-CV-SPC-0001.

Concrete footpaths require no base course and therefore shall be laid directly onto compacted sub-grade.

In installing bitumen sealed footpaths a base course consisting of compacted crushed rock of at least 75 mm thickness shall be laid. In addition the base course shall extend 200 mm beyond the edge of the seal.

5.14.2 Concrete Footpaths

All concrete works shall be in accordance with Ma'aden Specification MD-101-SMEM-EG-ST-SPC-0005 Concrete Works.

The concrete used for footpaths shall conform to ACI 301-10 and ACI 304R-00, be grade N25 with a specified slump of 80 mm.

Unless specified otherwise, the thickness of the footpaths shall be a nominal 75 mm.

Contraction and expansion joints shall be provided in accordance with the drawings and the following specifications:

- In the event of an expansion and contraction joint coinciding, the expansion joint shall be provided
- Contraction joints shall be provided at 1.5 m centres and shall have a smooth trowel finish for 80 mm either side of the joint
- Expansion joints shall be provided at 6.0 m centers and shall consist of a 12 mm gap filled with an approved 12 mm thick bitumen impregnated filler board for the full depth of the concrete
- The Contractor shall ensure that the filler board finishes flush with the surface of the path
- Prior to installing the proposed filler board, the Contractor shall submit details to the Employer. The joints shall have a smooth trowel finish for 80 mm either side of the joint.

The surface of the concrete footpath shall have a broom finish.

5.14.3 Asphalt Sealed Footpaths

Asphalt sealed footpaths shall consist of a single coat of prime/binder covered with 5 mm aggregate. The footpath shall be constructed in accordance with Sections 5.14.1and 5.14.2.

5.15 Road Crossings

Ducts for service crossings shall be supplied and installed in accordance with the drawings.

Earthworks for road crossings and services shall conform to Ma'aden Earthworks specification MD-101-SMEM-EG-CV-SPC-0001.

Backfilling over all ducts shall be sand, compacted in layers of 150 mm maximum thickness. The sand shall be compacted to at least 95% of the maximum dry density obtained in modified maximum dry density compaction tests in accordance with ASTM D2487.

5.15.1 Installation

The service ducts shall be placed at 90° to the frontage of the lot and shall be installed prior to the commencement of the road surfacing works.

The end of the duct servicing the lot shall be set within 500 mm of the front boundary and within 500 mm of the side boundary of the adjacent lot. The opposite end of the duct shall terminate within 500 mm of the alignment of the proposed main.

The service ducts shall be set at a minimum of 100 mm clearance between adjacent ducts and be allowed a maximum deviation in alignment of 100 mm.

Prior to backfilling, the end of the service duct shall be wrapped with identification tape that shall be brought to the surface.

On completion of the Curbing the duct shall be indicated by placing a blue painted full triangle of 100 mm sides, apex down, directly above each end of the duct. The paint shall be suitable for applying to Curb material and weather resistant.

5.16 Verge and Medians

Verge and medians shall be backfilled in accordance with the drawings. In addition, unless noted otherwise, the final layer shall be topsoil or fine crushed rock from the top of the Curb to the road reserve boundary.

The topsoil shall be stabilized by hydro mulching, which shall be applied at a rate of 40 to 50 km per hectare by a pressurized spray. Typical proportions for the hydro mulch mixture are 1200 kg to 1400 kg newspaper, 200 L of pacifier and 10,000 L of water.

5.17 Edge Guideposts, Road Markings and Delineators

Guideposts shall be installed in accordance with the drawings. The posts shall be of an approved timber and free from bark.

The posts shall be 1.5 m long with a cross sectional dimension of 90 mm x 40 mm. They shall be painted white and fitted with delineators, being conic cube reflectors designed in accordance with the local authority's requirement and MUTCD (Manual on Uniform Traffic Control Devices).

Each post shall be fitted with a red and white delineator, fixed with screws and located 130 mm down from the top of the post on the centerline of the 90 mm face. The red delineator shall be on the left and the white delineator on the right when viewed in the direction of travel.

The posts shall be embedded 500 mm into the ground.

5.18 Signs

The signs shall be located as shown on the drawings and the Works shall also include the removal and reinstatement of existing signs where required.

Unless noted otherwise street name signs shall be installed at the ends of each new street and at the intersection with other streets.

The traffic signs and associated hardware shall be manufactured in accordance with the local authority's requirements and MUTCD.

5.19 Roadway Marking and Signs

All roadway markings and signs shall be fabricated and installed as per the drawings approved by the Employer.

6. INSPECTION

6.1 General

The Employer will inspect the Work while in progress, as well as at the completion of various operations by the Contractor whose role may include the following:

- To inspect and monitor foundation surface preparation prior to placement of any fill materials including aggregates, this may involve being presented with the performance of field and laboratory tests
- Monitor in-place density and moisture content tests on compacted fill to verify conformance to this specification and any other field or laboratory testing as may be required
- To inspect and monitor surface preparation prior to placement of any concrete and/or asphalt, this may involve being presented with the performance of field and laboratory tests.

6.1.1 Tolerances

6.1.1.1 Sub-Grade

Finished sub grade beneath pavement shall be prepared to a tolerance of +5 mm/-35 mm from the design levels shown on the drawings.

6.1.1.2 Pavement and Seal Width

The width of the pavement and primer seal shall be such that the dimensions measured at right angles from the designated centerline to either edge of the pavement or seal is greater than, or equal to, the dimension shown on the drawings.

6.1.1.3 Grades

The upper surface of each course shall be tested by a straight edge to ensure accuracy and any irregularities greater than the tolerances specified elsewhere herein shall be made good by the addition or removal of material and further rolling until the cross section shown and/or specified is obtained. Where additional material is required, the surface shall be reworked and compacted in accordance with the requirements of this Specification and trimmed to the tolerances allowed.

The finished levels of the respective layers shall be prepared to the tolerances detailed in Table 6-1.

Table 6-1 : Layer Tolerances

Tolerance for:	**Heavily Trafficked Roads** (as defined in the project documentation)	**Other Roads**
Sub base surface versus design levels	+5 mm/-25 mm	+5 mm/-25 mm
Base Course surface versus design levels	+5 mm/-15 mm	+5 mm/-15 mm
Surface Course surface versus design levels	+5 mm/-5 mm	+5 mm/-10 mm
Deviation from a 3 m straight edge (any direction) Base Course Surfacing	 10 mm 3 mm Longitudinal, 5 mm Transverse	 10 mm 5 mm either direction

In addition to the above, any deviations of the surfacing from design levels shall not cause ponding on the surface.

6.1.1.4 *Cross Falls*

The cross falls on the finished pavement shall not vary by more than 0.5% from the cross falls shown on the drawings.

7. TESTING

Samples required for testing by this specification shall be furnished and delivered as directed by the Employer.

A test report from a certified testing laboratory on all materials proposed for use shall be submitted for written approval by the Employer prior to use of any materials. Certificates shall be provided with each batch of materials delivered to the jobsite.

Substitute materials may be offered for approval. Required samples, data, test, or information shall be furnished.

It shall be the responsibility of the Contractor to accomplish the requirements contained in this Specification.

Compliance tests for specified compaction shall be in accordance with the following:

- Maximum density and optimum moisture content shall be determined in accordance with ASTM D1557 Method D
- In-place density shall be determined in accordance with ASTM D1556, or D6938. When ASTM D6938 is used, the nuclear Densimeter shall be calibrated by comparison with results from ASTM D1556. Initially, three check tests of density and moisture shall be performed by the sand cone method. Thereafter, one test in ten shall be verified with sand cone test. At least one check shall be performed each day for each material. Nuclear Densimeters shall be used under freezing conditions.

- Field in-place density tests will be made, as required, to verify that the specified compaction is achieved. Unless otherwise directed by the Employer's Representative, a minimum of one test per shift will be made in each different type of fill material being placed, and a minimum of one test per 500 m^2 of compacted aggregate base course and surface course.
- Where granular soils with less than 5% of material passing the No. 200 sieve are used, relative density shall be determined in accordance with ASTM D4253 and D4254.

All other tests shall be as prescribed in this Specification or the standards referenced therein.

Where work or materials do not meet the specified requirements, these shall be reworked and/or replaced until it complies with the specified requirements.

8. SUBMITTALS

- Certificates of compliance to the applicable standard specifications shall be submitted in accordance with the contract or as requested by the Employer
- Contractor shall provide a Quality Assurance and Control Plan for the control quality of items and services to meet the requirements of this specification and applicable codes and standards. The plan shall include, as a minimum, the following to cover all design and construction operations, both onsite and offsite:
  - Name, qualifications, duties responsibilities and authorities of the Contractors Field Quality Control Representative (FQCR)
  - Control, verification and acceptance testing procedures
  - Reporting procedures.

9. QUALITY CONTROL

The Contractor shall refer to MD-101-SMPM-PM-QA-PEI-0001 "Contractor Quality Requirements".